CN 09.04.2025

(lab 7-S7)

MIPS (million instruction / sec)

$$MIPS = \frac{IC}{CPU_{TIME} \times 10^6} = \frac{IC}{\frac{IC \times CPI}{CR} \times 10^6} = \frac{CR}{CPI \times 10^6}$$

MFLOPS (million FLOPS / sec)

$$MFLOPS = \frac{FLOPS}{EXECUTION\ TIME \times 10^6}$$

$$speedup = \frac{CPU_{TIME\ old}}{CPU_{TIME\ new}};$$

speedup = A

$$CPU_{TIME\ new} = \frac{portion\ enhanced}{speedup} + CPU_{TIME\ unenhanced}$$

$$CPU_{TIME\ all} = CPU_{TIME\ ALU} + CPU_{TIME\ L/S}$$

$CPU_{TIME} = 100s$

multiply ops take 80% of the time

How much do I improve multiply performance to run the program 5 times faster?

$$20s = \frac{CPU_{TIME\ mult}}{x} + (100 - CPU_{TIME\ mult})$$

$$20 = \frac{80}{x} + 20$$

! NU SE POATE

$$\Rightarrow speedup = \frac{1}{\frac{f}{s} + (1-f)}$$

Amdahl' Law

f - portion enhanced

1.9.

	CPi	IC
ALU	1	2.56 E9
L/S	12	1.28 E9
BR	5	$256 \cdot 10^6$

CR = 2 GHz

ALU and L/S IC divided by 0.7 / core

a) ALU_{time} 1,2,4 - cores ; speed up for 2,4 -cores relative to 1

b) if CPI_{ALU} is 2x, impact on CPU_{time} 1,2 cores

c) to what would $CPI_{L/S}$ be reduced for 1-core par to match 4-core with initial CPi

a) $\sum_{i=1}^{3} CPI_i \times IC_i = 1 \cdot 2{,}56 \cdot 10^9 + 12 \cdot 1{,}28 \cdot 10^9 + 5 \cdot 0{,}256 \cdot 10^9 = 15 \cdot 1{,}28 \cdot 10^9$

$$CPU_{TIME} = \frac{CC}{CR} = \frac{15 \cdot 1{,}28 \cdot 10^9}{2 \cdot 10^9} = 9{,}6\,S - 1 \text{ core}$$

$$CPU_{TIME_2} = \frac{11 \cdot 1{,}28 \cdot 10^9}{2 \cdot 10^9} = 7{,}04\,S$$

$$CC_2 = \frac{1}{1{,}4}\left(1 \cdot 2{,}56 \cdot 10^9 + 12 \cdot 1{,}28 \cdot 10^9\right) + 5 \cdot 0{,}256 \cdot 10^9 =$$

$2 \cdot 0{,}7$

$$= 11 \cdot 1{,}28 \cdot 10^9$$

$$CC_4 = \frac{1}{2{,}8}\left(1 \cdot 2{,}56 \cdot 10^9 + 12 \cdot 1{,}28 \cdot 10^9\right) + 5 \cdot 0{,}256 \cdot 10^9 =$$

$$= 6 \cdot 1{,}28 \cdot 10^9$$

$$CPU_{time_4} = \frac{6 \cdot 1{,}28 \cdot 10^9}{2 \cdot 10^9} = 3{,}84\,s$$

$$speedup_2 = \frac{CPU\ time_1}{CPU\ time_2} = \frac{9,6}{7,04} = 1,36$$

$$speedup_4 = \frac{CPU\ time_1}{CPU\ time_4} = \frac{9,6}{3,84} = 2,5$$

b) $CPI_{ALU} = 2$

$$CC_1 = 2 \cdot 2,56 \cdot 10^9 + 12 \cdot 1,28 \cdot 10^9 + 5 \cdot 0,256 \cdot 10^9 = 17 \cdot 1,28 \cdot 10^9$$

$$CPU_{time_1} = \frac{17 \cdot 1,28 \cdot 10^9}{2 \cdot 10^9} = 10,88$$

$$CPU_{time\ new} = \frac{CPU_{time\ ALU}}{\frac{1}{2}} + CPU_{time_2\ all} - CPU_{time\ ALU} =$$

$$= CPU_{time\ ALU} + CPU_{time_2\ old} =$$

$$= \frac{\frac{1}{1,4} \cdot 1 \cdot 2,56 \cdot 10^9}{2 \cdot 10^9} + 7,04\ S$$

$$= \frac{1,28}{1,4}\ S + 7,04\ S =$$

$$= 7,95\ S$$

$$speedup_1 = \frac{CPU\ time_1\ old}{CPU\ time_1\ new} = \frac{9,6\ S}{10,88\ S} = 0,88$$

$$speedup_2 = \frac{CPU\ time_2\ old}{CPU\ time_2\ new} = \frac{7,04}{7,95} = 0,89$$

c) $CPU\ time_{1\ new} = CPU_{time_4} = 3,84\ S$

2.

	CR	CPi	IC
P1	4 GHz	0,9	5.0E9
P2	3 GHz	0,75	1.0 E9

a) compare performances

$$CPU_{time\ 1} = \frac{ic_1 \times CPi_1}{CR_1} = \frac{5 \cdot 10^9 \cdot 0{,}9}{4 \cdot 10^9} = \frac{4{,}5}{4} = 1{,}1\,S$$

$$CPU_{time_2} = \frac{ic_2 \times CPi_2}{CR_2} = \frac{1 \cdot 10^9 \cdot 0{,}75}{3 \cdot 10^9} = \frac{0{,}75}{3} = 0{,}25\,S$$

b) if P_1 executes 1.0 E9 inst, $IC_{P_2} - CPU_{time\ P_1} = CPU_{time\ P_2}$

$$ic_{1\ new} = 1.0E9$$

$$CPU_{time\ 1\ new} = \frac{1 \cdot 10^9 \cdot 0{,}9}{4 \cdot 10^9} = \frac{0{,}9}{4} = 0{,}23S$$